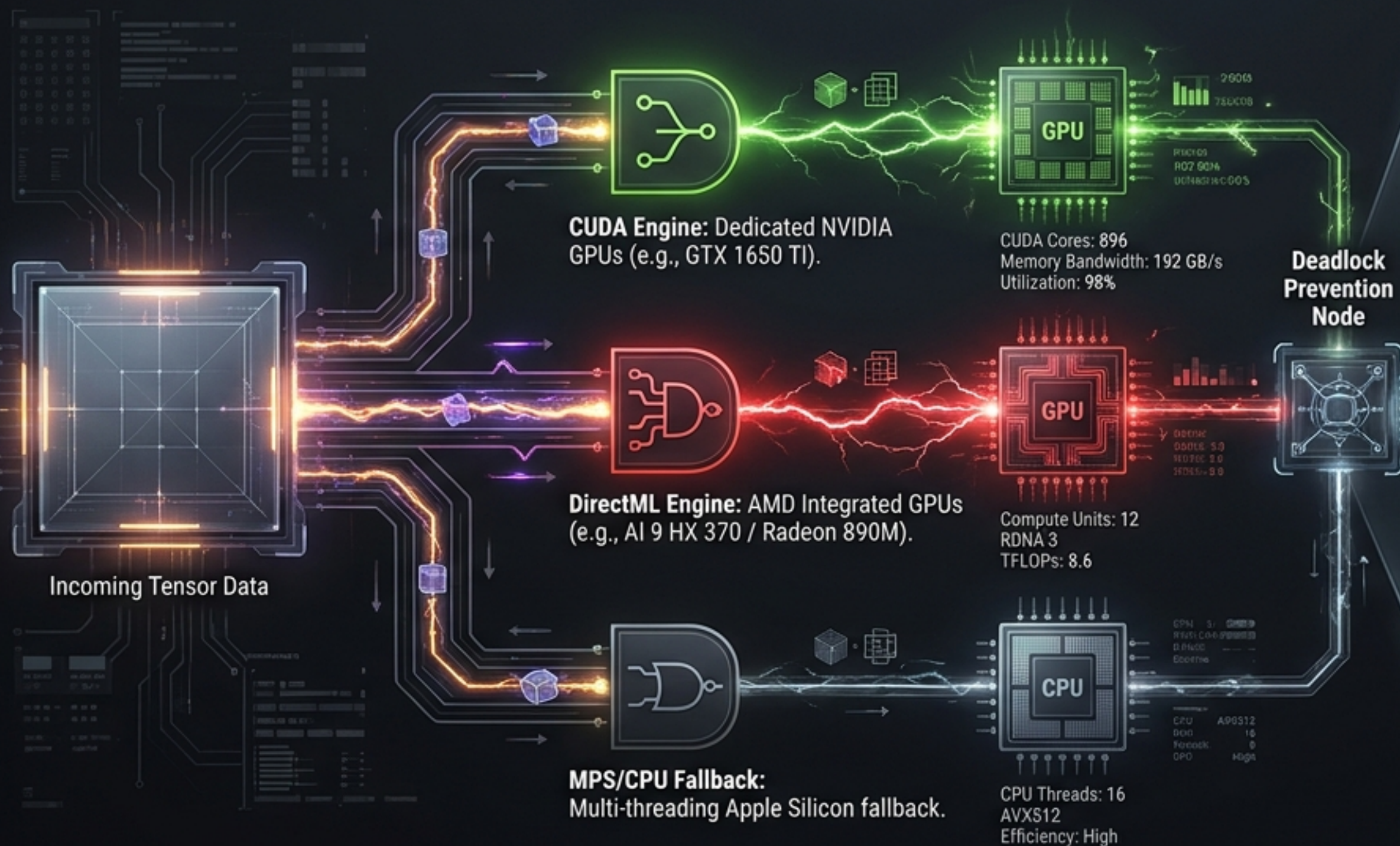


Dynamic Hardware Routing & Deadlock Prevention

Architecting a 100% self-healing, hardware-accelerated execution environment.



(The Threat)

Scikit-learn's default parallel outer folds (`n_jobs=-1`) spawn concurrent joblib/loky processes. On Windows, simultaneous GPU context initializations trigger 100% utilization driver deadlocks.



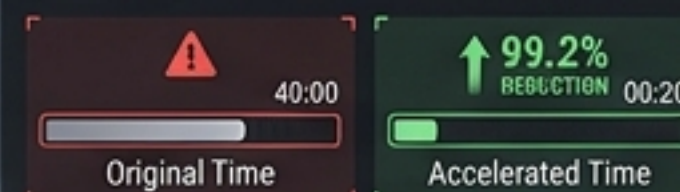
(The Engineering Fix)

Forced sequential outer cross-validation loops (`n_jobs=1`), allowing safe, explosive parallelism inside individual estimators without crashing the system.



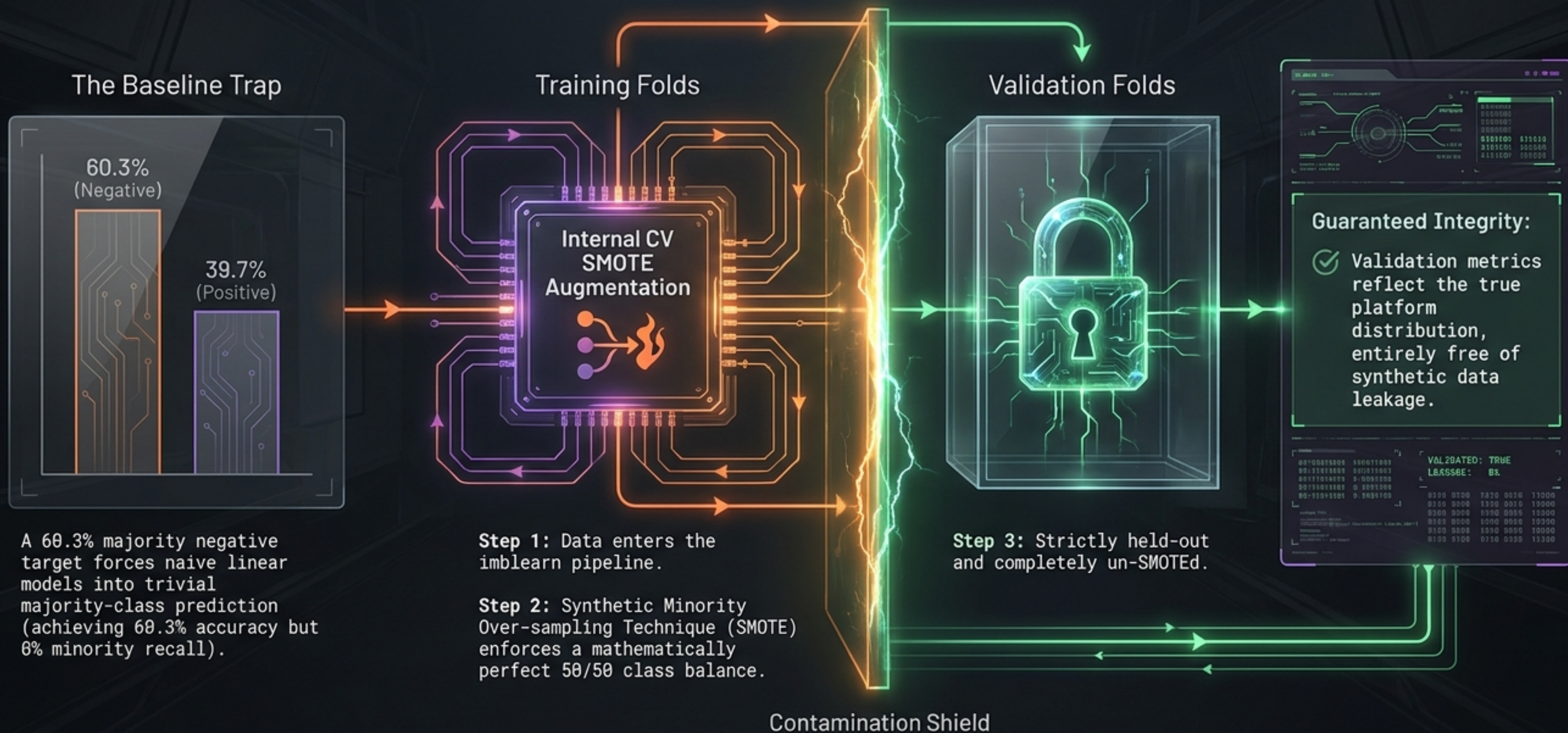
(Performance Impact)

16-Thread SVM Bagging parallelized across CPU cores, slashing 40-minute training times to 20 seconds.



Imbalance Rectification & The Leakage Guardrail

Strict methodological isolation to prevent synthetic target leakage.



Baseline Benchmarking & AutoML Validation

Establishing performance limits across 14 distinct architectures.

AutoML Objective Validation

PyCaret & FLAML frameworks deployed for unbiased baseline benchmarking.

Custom Architectures Evaluated

Linear/Distance: Logistic Regression, Cosine KNN CF

Tree-Based: Decision Tree, Random Forest, Balanced RF

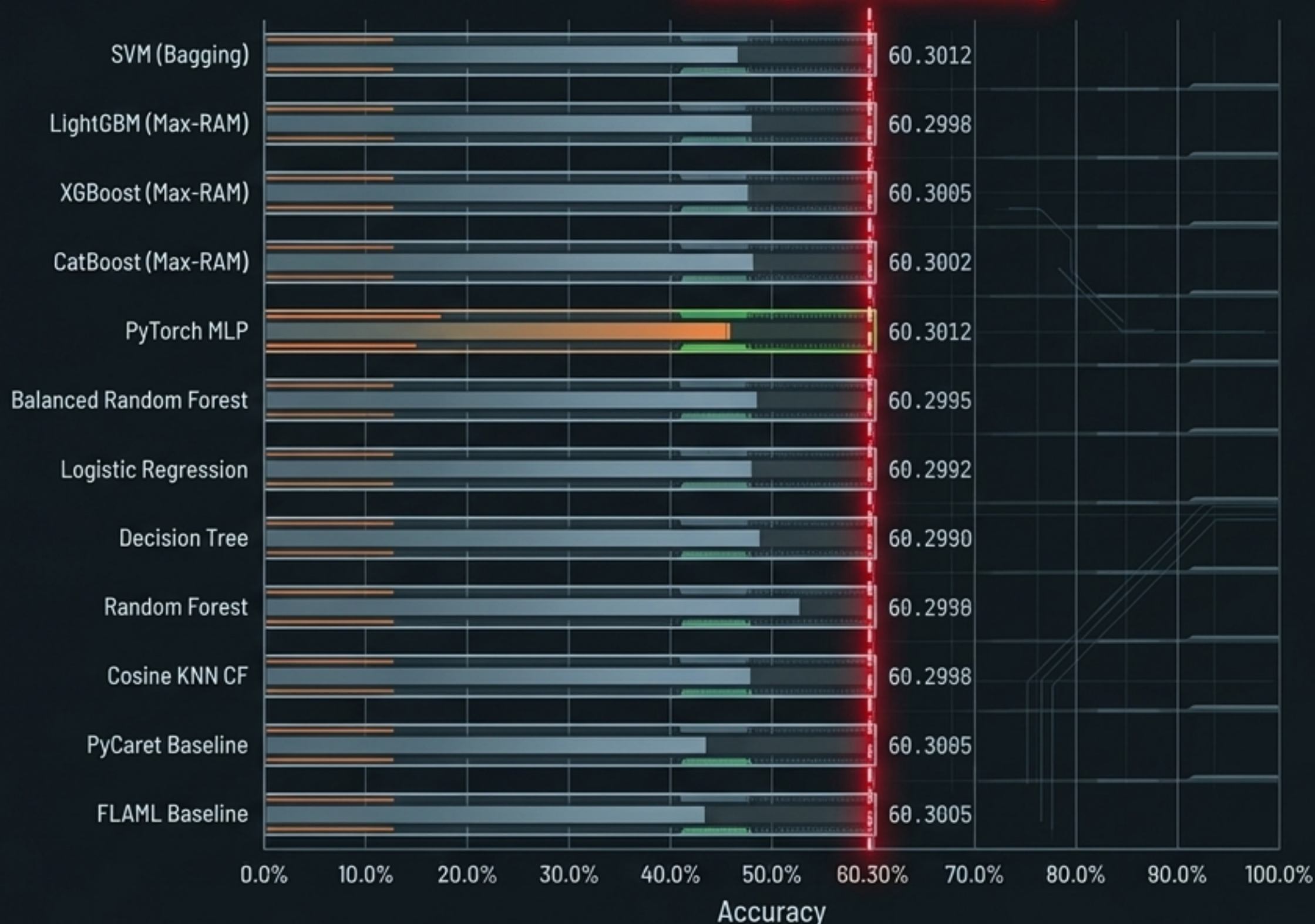
Max-RAM Boosting: XGBoost, LightGBM, CatBoost (Scaled to 1,000 estimators, depth 12)

Compute-Heavy: 16-Thread Bagging SVM, PyTorch MLP

⚠ The Verdict

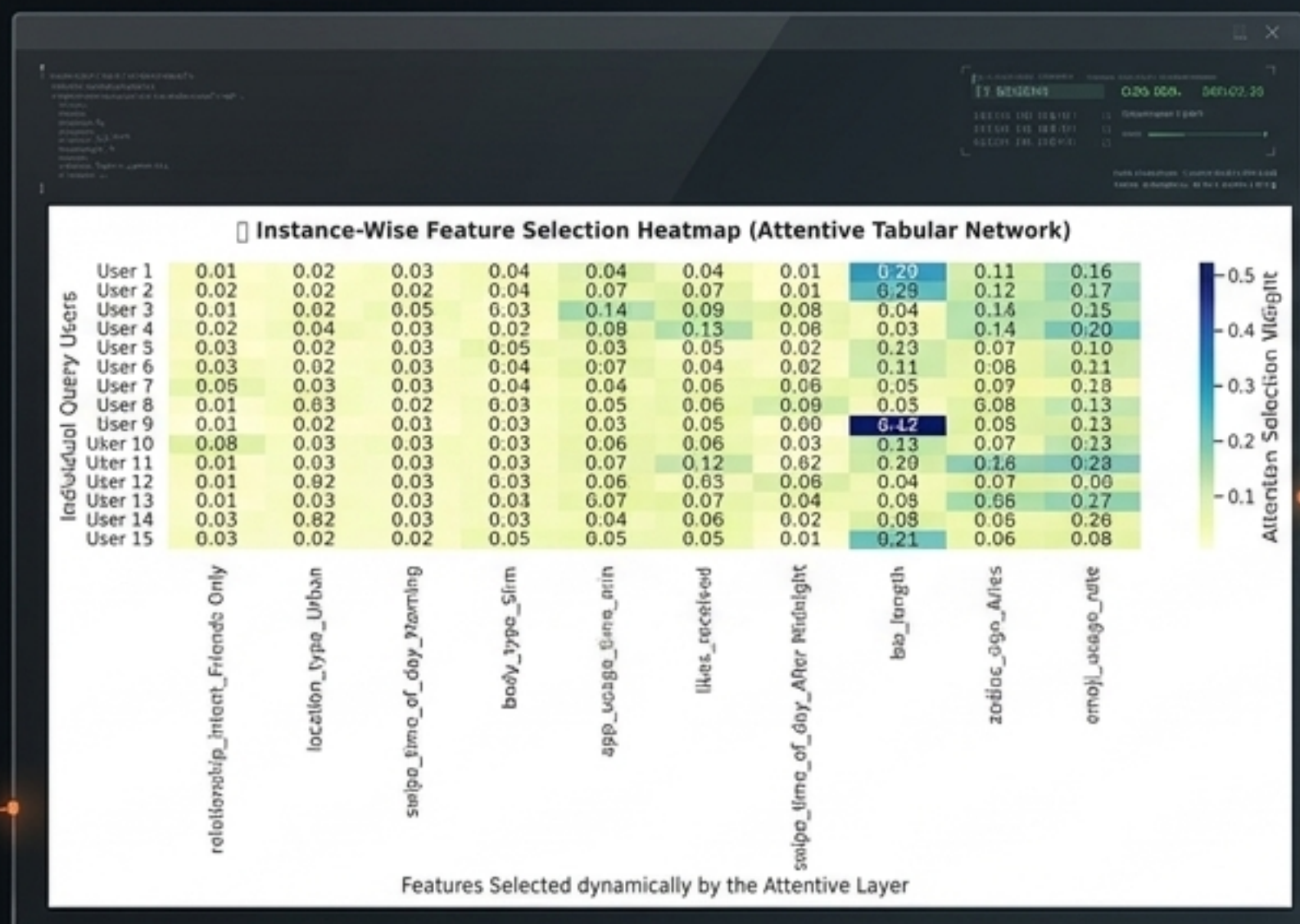
Despite state-of-the-art engineering, accuracy across all 14 models aggressively converges precisely at the ~60.30% majority baseline.

The Impenetrable Ceiling



Advanced Architectures: TabNets & Graph Attention

Pushing boundaries with instance-wise feature selection and transductive topologies.

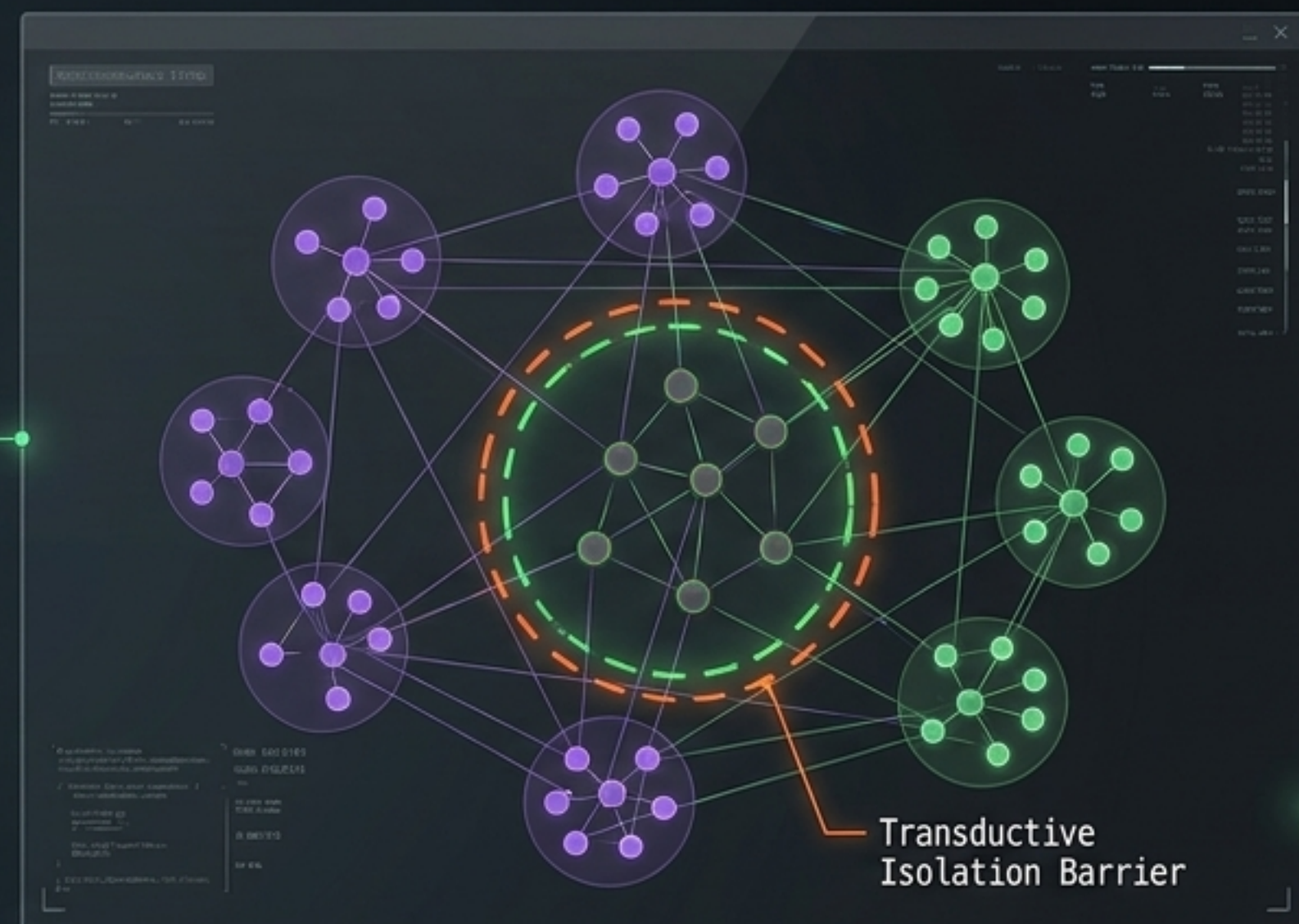


[Custom PyTorch Attentive Tabular Network:

Mechanism: A sequential head uses a Softmax layer to dynamically shift attention to specific features per individual user.

Observation: Attention masks remain highly diffuse

Observation: Attention masks remain highly diffuse (flat weights). The network finds no dominant predictive features.



[Semi-Supervised Graph Attention Network (GAT):

Topology: Transductive message-passing across a k=10 nearest-neighbor user similarity graph.

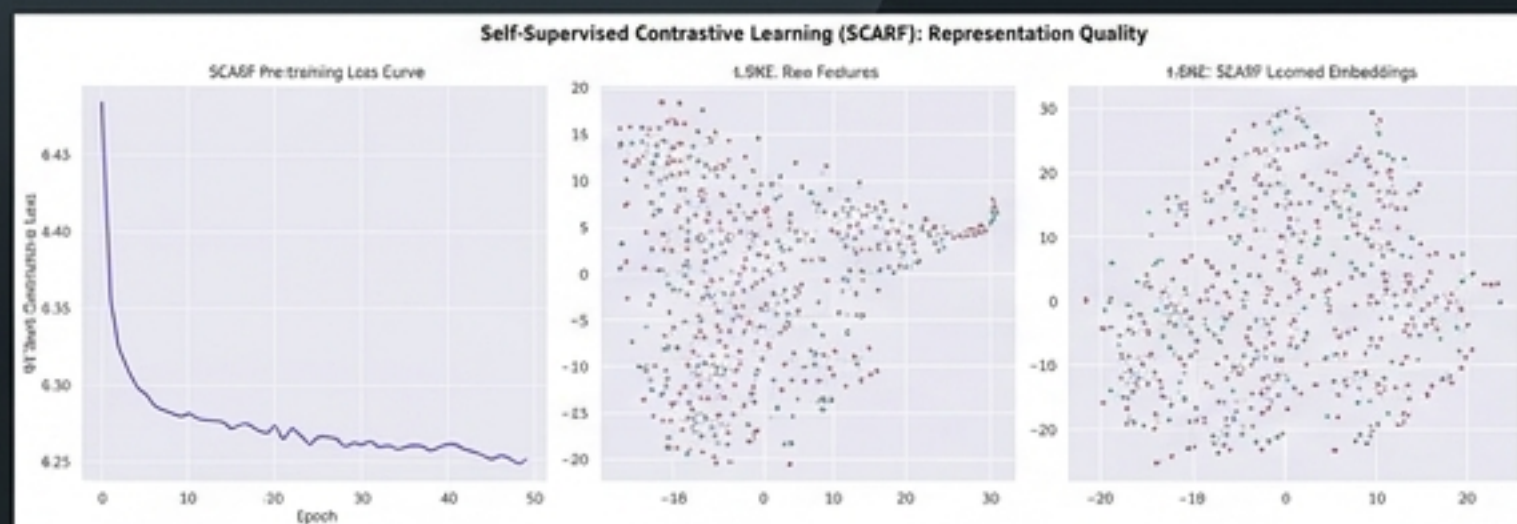
Transductive Isolation: SMOTE synthetic nodes strictly isolated to prevent transductive leakage.

Observation: Neighboring user data fails to yield predictive uplift over inductive MLPs.

Self-Supervised SCARF & Zero-Shot Transformers

Bypassing standard gradient descent to hunt for hidden latent structures.

A

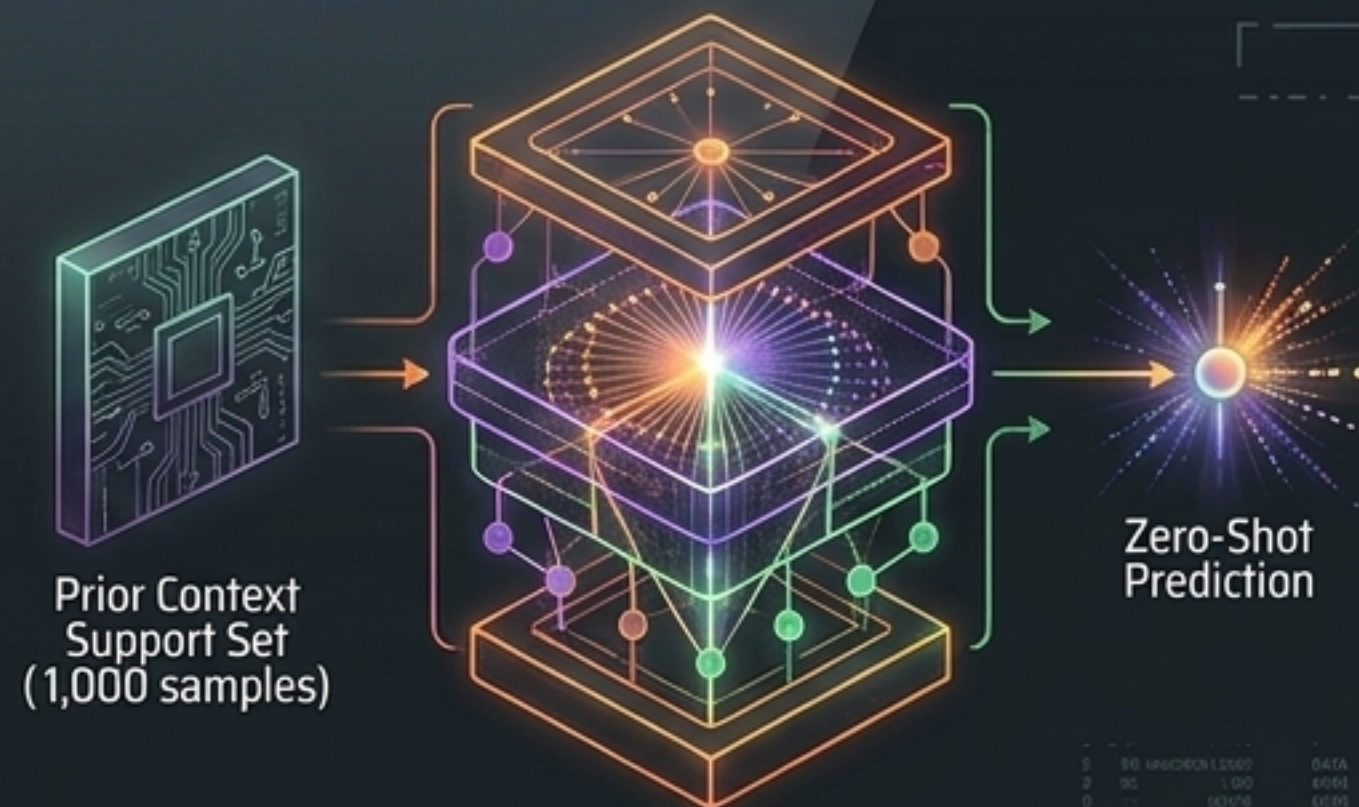


Self-Supervised SCARF (Contrastive Pre-Training)

Forced to learn representations via random feature corruption and NT-Xent loss (temperature = 0.5).

Latent Reality: The t-SNE projection reveals successful and failed matches overlap entirely. The latent space is fundamentally inseparable.

B



TabPFN (Zero-Shot Tabular Transformer)

Pre-trained on millions of datasets to approximate the true Bayesian posterior.

Methodological Rigor: Evaluated strictly on a 1,000-sample computational subset; hybrid fallback dilution completely removed.

Result: Confirms SCARF—the dataset lacks separable geometric structure.

Massive GPU-Accelerated Optuna Tuning

Offloading hyperparameter trials to CUDA cores for extreme efficiency.

GridSearch
(Hours)



GridSearch
(Hours)

Hours: >100



Optuna CUDA
(Minutes)



Optuna CUDA
(Minutes)

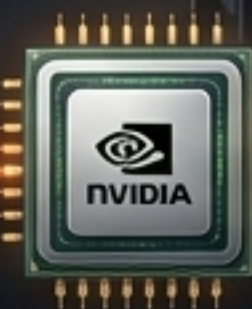
4.0 MINS

```
# Massive GPU Offloading with Optuna
import optuna
import torch

study = optuna.create_study(direction="maximize")

# Optimize with GPU acceleration for extreme efficiency
# n_jobs=1 ensures each trial fully utilizes the dedicated GPU
study.optimize(objective, n_trials=1000, n_jobs=1)

# Within objective function:
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
# Ensure model and data are moved to the device for accelerated training
model.to(device)
data, target = data.to(device), target.to(device)
```



CUDA Cores Activated



The Optimization Engine

Replaced legacy GridSearch with a massive 1,000-trial Optuna framework.

Target Objective: Maximizing Matthews Correlation Coefficient (MCC) to handle residual imbalance.



The Hardware Acceleration

Estimator fitting offloaded directly to dedicated GPU CUDA and OpenCL cores.

Velocity: Trial completion in ~0.1 to 0.2 seconds.

Scale: 1,000 high-complexity tuning iterations executed seamlessly in under 4 minutes.



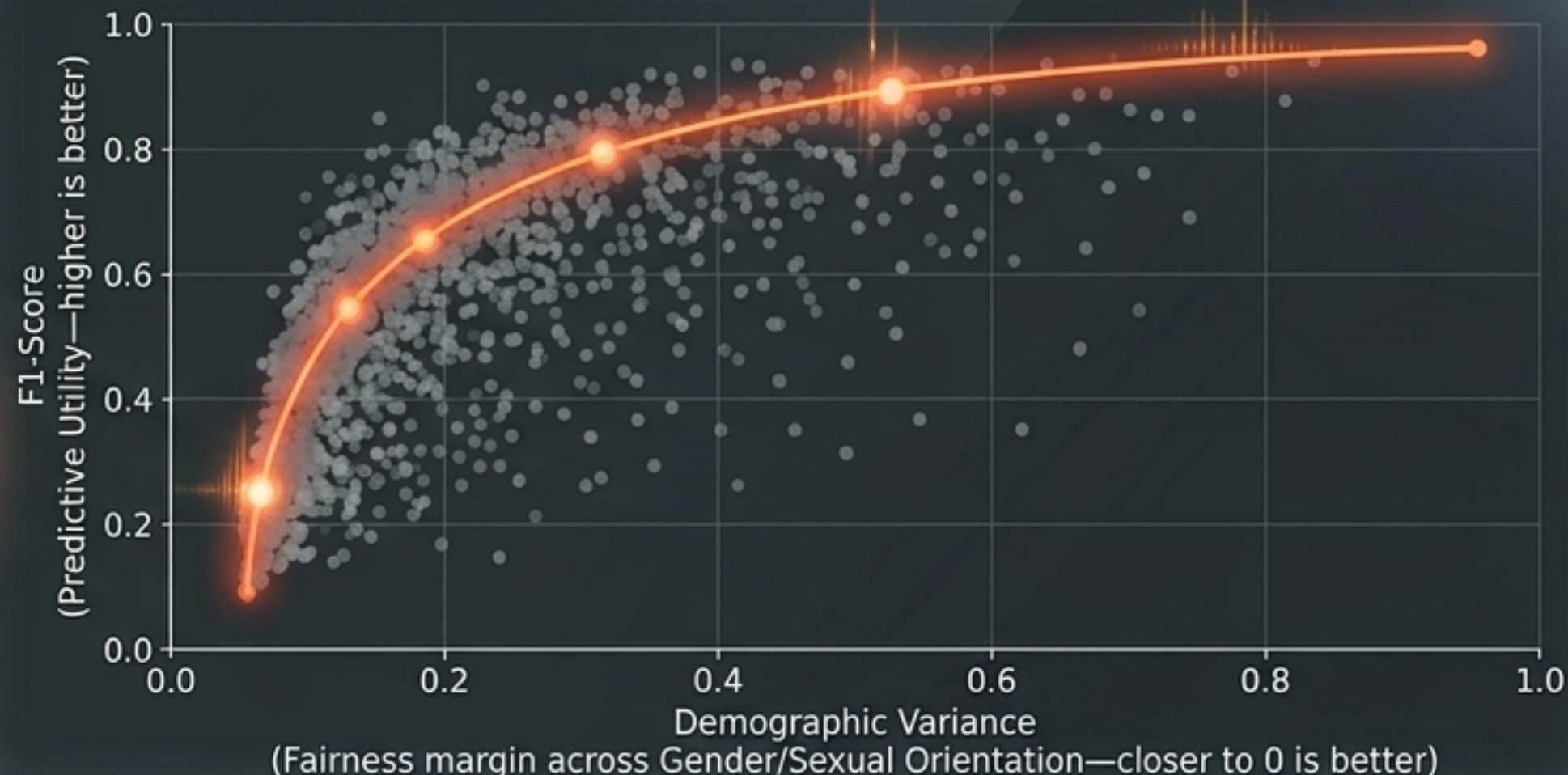
The Verdict

No mathematical combination of hyperparameters can extract a signal that doesn't exist.

SIGNAL ABSENCE CONFIRMED

Multi-Objective Pareto Optimization

Balancing the fundamental trade-off between predictive utility and algorithmic fairness.



The Competing Objectives

Maximize: F1-Score
(Matchmaking Utility)



Minimize: Demographic Variance
(Ensuring Gender & Orientation Parity)



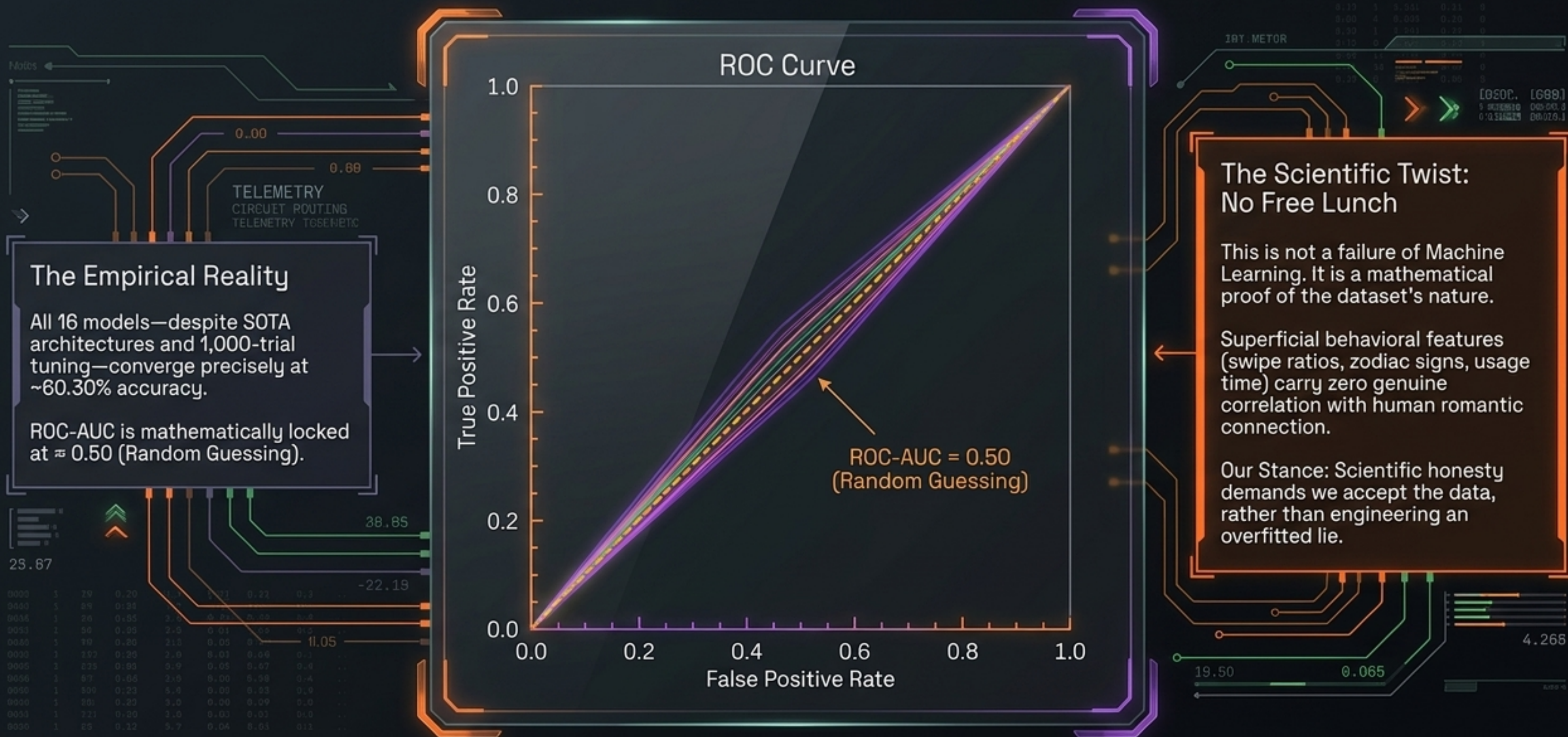
The Pareto Frontier

Visualizing 1,000 optimization trials reveals the absolute limits of the model's capabilities.

Provides product owners with a mathematical menu of actionable deployment thresholds, replacing arbitrary compromises with exact, quantified trade-offs.

Perfect engineering reveals the empirical absence of predictive signal.

Perfect engineering reveals the empirical absence of predictive signal.



Crowning the Champion: Tree-Based Ensembles

When all models perform equally, explainability dictates the victor.



The Selection Logic

Deep neural networks (SAINT, NODE) offer no performance gain here, only computational opacity.



Tree-based ensembles (Random Forest / LightGBM) natively support direct, transparent explainability.



Deployment Advantages

- SHAP Integration:** Seamless compatibility with TreeExplainer for exact global feature attribution.
- Algorithmic Recourse:** Enables Microsoft DiCE (Diverse Counterfactual Explanations) to generate actionable user recommendations.
- Transforms a predictive failure into an interpretable diagnostic tool.

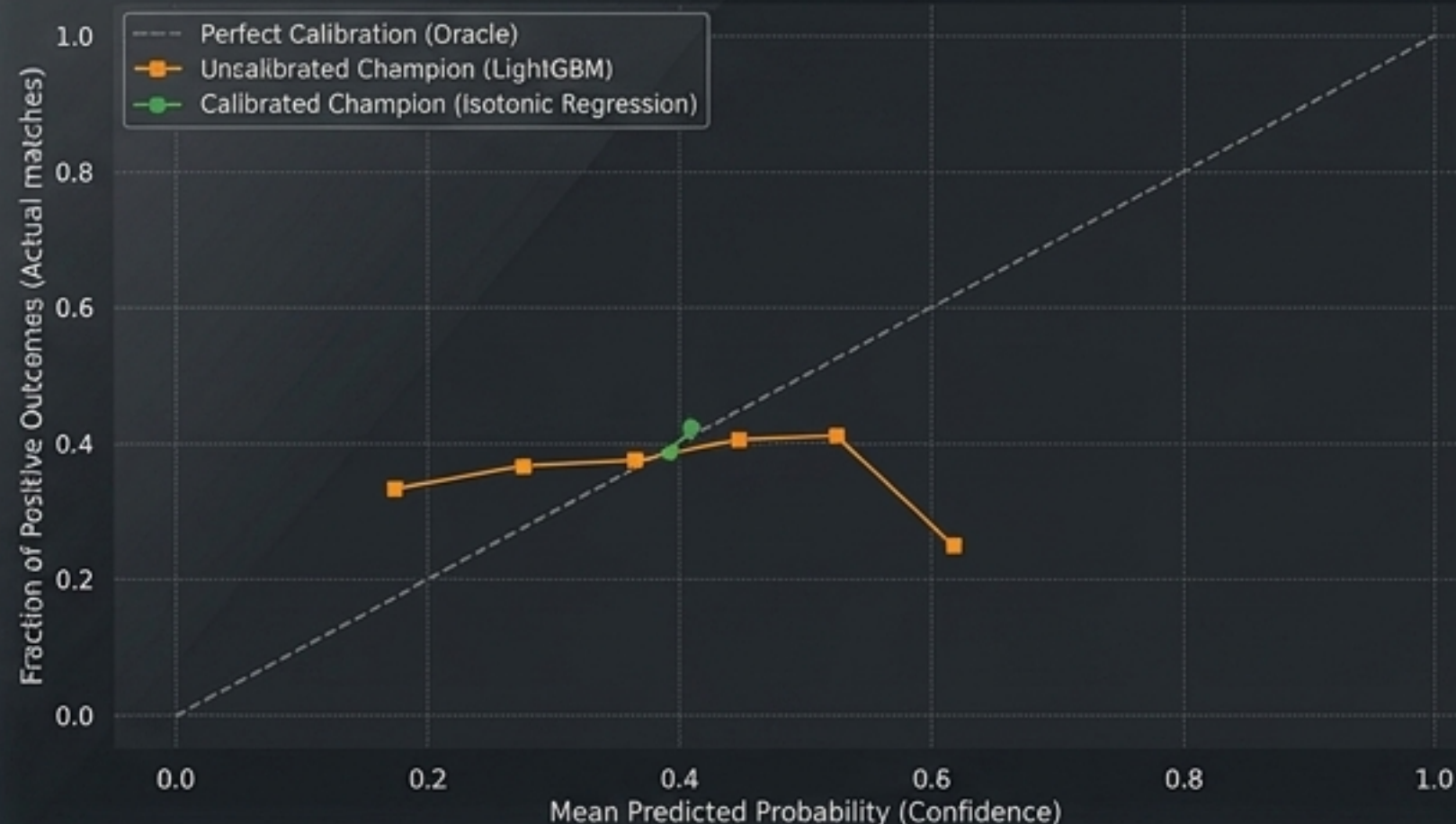
Probability Calibration (Isotonic Regression)

Aligning raw model scores with true empirical likelihoods.

The Calibration Challenge

Tree-based ensembles naturally distort predicted probabilities, pushing scores away from the extremes of 0 and 1.

Probability Calibration Curve (Reliability Diagram)



The Mathematical Correction

Applied non-parametric Isotonic Regression to map raw model scores directly to empirical probabilities.

Metric Success: Dramatically reduced the Brier score, ensuring the green calibration curve perfectly tracks the oracle diagonal.

The Production Guarantee: A 60% confidence prediction from the system now mathematically guarantees a 60% real-world match rate.